Hein Zarni Aung

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EDUCATION

Stony Brook University Expected Graduation: May 2027

B.S., Computer Science + Applied Math & Statistics (Data Science) | GPA: 3.40

Honors & Scholarships: Global Excellence Scholarship

Relevant Coursework: Full Stack Development, Statistics & Probability, Software Security

INTERNSHIPS

Software Engineering Intern — Clustr Jan 2026 – Sept 2026 (expected)

Cross-platform React Native social/scheduling app | ~8-person dev team

- Built server-side community infrastructure using Firebase Cloud Functions, including CRUD operations for communities, posts, and comments with enforced ban/edit permissions to prevent client-side database tampering, plus join/leave workflows and activity feed endpoints.
- Designed and implemented a Scheduler UI module for the React Native mobile app, featuring tabbed navigation (Suggestions, Requests, Responses), a reschedule bottom sheet, and custom date/time picker integration to improve user engagement.

Software Engineering Intern — Prenora Dynamics LLC Jun 2026 – Sept 2026 (expected)

~8-person team

- Own end-to-end design decisions for an in-house, in-app video chat feature, researching and evaluating available SDKs/approaches to replace reliance on third-party platforms (e.g., Zoom, Google Meet).
- Set up the PostgreSQL/Supabase database and backend data models for Jamory, an in-house marketplace app connecting students with music tutors, and built a seed-data workflow that let the rest of the team build and test independently against the database.

RESEARCH

Undergraduate Research Assistant — Ethos Lab, Dept. of Computer Science, Stony Brook University Sep 2025 – Dec 2025

PI: Abisheka Pitumpe | Team: 2

- Built a pipeline to monitor newly issued SSL/TLS certificates in real time and flag domains impersonating EZ-Pass tolling services, using Selenium to scrape site data and Python-based heuristics to triage flagged domains for fraud/security review.

CAMPUS INVOLVEMENT

Teaching Assistant – Data Structures & Algorithms, Dept. of Computer Science, Stony Brook University Aug 2025 – Dec 2025

- Created original PowerPoint-based review materials from scratch for midterm exam preparation; materials were adopted as the basis for subsequent semesters' review sessions.
- Hosted 2-hour midterm review sessions reaching 100+ students combined in-person (~50) and via Zoom, translating complex Data Structures & Algorithms concepts for a large, mixed-experience audience.

PROJECTS

TuneAcademy React, NumPy, Essentia, FastAPI, GitHub Actions

- Built the AI/audio-analysis pipeline for an app matching music students with instructors based on specific technical weaknesses, using Essentia and NumPy to extract pitch, rhythm, and tone features from recorded performances.
- Set up a CI/CD pipeline using GitHub Actions to lint, test, and auto-deploy the app to Cloudflare Workers, and integrated ElevenLabs' generative AI voice API to produce spoken feedback reports for students.

Diary Handwriting Analysis Python, PyTorch, Hugging Face Transformers, OpenCV

- Fine-tuned Microsoft's TrOCR handwriting recognition model with a custom OpenCV line-segmentation preprocessing pipeline (grayscale conversion, binary thresholding, horizontal ink-projection profiling), reaching ~87% reliable page transcription on personal diary pages.

SKILLS

Languages: Python, JavaScript, TypeScript, Java, C

AI/ML: PyTorch, Hugging Face Transformers, OpenCV, ElevenLabs (GenAI API)

Frontend: React, React Native, Next.js

Backend: Node.js, Express.js, FastAPI

Databases & Cloud: MongoDB, PostgreSQL, Firebase/Firestore, AWS, GCP, Supabase

Tools: Git, Docker, Linux, PM2, Selenium